FROB C5 finite-field factor-kernel audit

Validation audit, 22 October 2026

Scope. Frozen `EXP-FROB-91ee9c` control C5; Lean source commit `2143dda2b698cf6e3fb4bafbe853af7262e3ffba`. These are mathematical statements about canonical degree-five finite fields. They do not replay a scientific trial or establish an ECDLP performance claim.

Base field	Nontrivial factors of $X^5 - 1$	Lean kernel result	Sum dimension
$\mathbf{F}_{11}$	$X + 2, X + 6, X + 7, X + 8$	four successive disjoint summands = $\ker \mathrm{Tr}$	4
$\mathbf{F}_{13}$	$X^4 + X^3 + X^2 + X + 1$	sole factor kernel = $\ker \mathrm{Tr}$	4
$\mathbf{F}_{19}$	$X^2 + 5X + 1, X^2 + 15X + 1$	disjoint sum = $\ker \mathrm{Tr}$	4

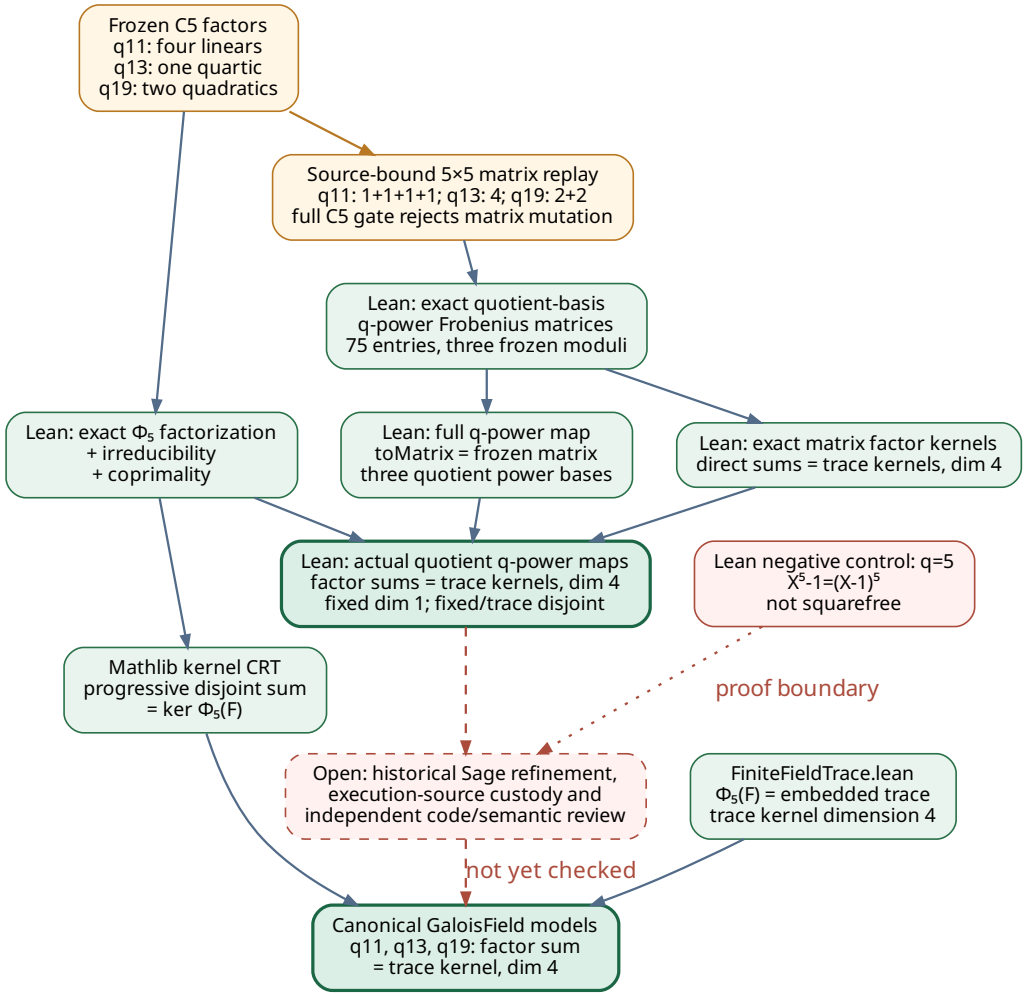
Proof method. Lean checks exact cyclotomic factorizations and irreducibility. For $q = 13, 19$, it uses Mathlib's finite-field cyclotomic factor criterion with $\mathrm{ord}_5(q) = 4, 2$; the $q = 11$ factors have degree one. Explicit coprimality certificates and Mathlib's polynomial-kernel CRT give the direct sums. The prior finite-field theorem identifies $1 + F + F^2 + F^3 + F^4$ with the embedded trace and proves its kernel has dimension four. Here F is the q-power Frobenius on `GaloisField q 5`.

Verification. The three exact-source verifier receipts accompany the report. They bind the target specifications, theorem source, pinned Lean 4.35.0-rc1 toolchain and dependency manifest. Each target axiom audit passed; the full 8,971-job build passed and 612 theorem declarations used standard axioms only. Independent semantic review remains pending.

Archived matrix replay. A separate plain-Python checker reads the three frozen outputs, recomputes each 5×5 Frobenius matrix from the recorded modulus, checks modulus/factor irreducibility and factor kernels, directness and trace equality. The same gates reject a mutated matrix in all three cells; the byte-exact receipt is included. A separate Lean certificate now proves all 75 q-power matrix entries in the frozen quotient bases. A second certificate proves direct factor-kernel sums and trace dimension four for the exact recorded matrices. A third Lean certificate identifies the full q-power quotient endomorphisms with those frozen matrices in their power bases. A fourth Lean certificate proves factor-kernel sums, trace dimension four, fifth-power and fixed-line controls for those actual quotient endomorphisms.

Negative control. A separate Lean theorem proves $X^5 - 1 = (X - 1)^5$ and non-squarefreeness over $\mathbf{F}_5$. This checks the coprimality boundary; $q = 5$ was not an archived core cell.

Boundary. Historical Sage code-path refinement, original execution-source custody and independent review remain open. The historical run control, C8 and experiment success gate therefore remain open. No measured value or performance dashboard curve changed.


Frozen C5 factors
q11: four linears
q13: one quartic
q19: two quadratics
Source-bound 5×5 matrix replay
q11: 1+1+1+1; q13: 4; q19: 2+2
full C5 gate rejects matrix mutation
Lean: exact quotient-basis
q-power Frobenius matrices
75 entries, three frozen moduli
Lean: exact Φ5 factorization
+ irreducibility
+ coprimality
Lean: full q-power map
toMatrix = frozen matrix
three quotient power bases
Lean: exact matrix factor kernels
direct sums = trace kernels, dim 4
Lean: actual quotient q-power maps
factor sums = trace kernels, dim 4
fixed dim 1; fixed/trace disjoint
Lean negative control: q=5
X5-1=(X-1)5
not squarefree
Mathlib kernel CRT
progressive disjoint sum
= ker Φ5(F)
proof boundary
Open: historical Sage refinement,
execution-source custody and
independent code/semantic review
FiniteFieldTrace.lean
Φ5(F) = embedded trace
trace kernel dimension 4
not yet checked
Canonical GaloisField models
q11, q13, q19: factor sum
= trace kernel, dim 4